

MAT 426, Class 23 Homework: Inner Product Spaces
(Due Tuesday, May 5, 2026)

1. Let $\mathcal{P}_2$ be the set of all polynomials of degree at most 2. Let $p(x) = a_0 + a_1x + a_2x^2$ and $q(x) = b_0 + b_1x + b_2x^2$.

(a) Show that $\langle p(x), q(x) \rangle = a_0b_0 + a_1b_1 + a_2b_2$ defines an inner product on $\mathcal{P}_2$.

(b) Let $p(x) = 3 - 2x$ and $q(x) = 1 + x + x^2$. Compute $\langle p(x), q(x) \rangle$.

2. Let a , b , and c be distinct real numbers. Show that

$$\langle p(x), q(x) \rangle = p(a)q(a) + p(b)q(b) + p(c)q(c)$$

defines an inner product on $\mathcal{P}_2$.

3. Determine which of the inner product axioms do not hold. Give a specific example in each case.

(a) Let $\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$. Define $\langle \mathbf{u}, \mathbf{v} \rangle = u_1v_1 - u_2v_2$.

(b) Let $\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$. Define $\langle \mathbf{u}, \mathbf{v} \rangle = u_1v_2 + u_2v_1$.

(c) In $\mathcal{P}_2$ define $\langle p(x), q(x) \rangle = p(1)q(1)$

(d) In M_{22} , define $\langle A, B \rangle = \det(AB)$.

4. Assume $\langle \mathbf{u}, \mathbf{v} \rangle$ is an inner product. Prove the following:

(a) $\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle$

(b) $\langle \mathbf{u}, c\mathbf{v} \rangle = c \langle \mathbf{u}, \mathbf{v} \rangle$

(c) $\langle \mathbf{u}, \mathbf{0} \rangle = \langle \mathbf{0}, \mathbf{v} \rangle = 0$

(d) $\langle \mathbf{u} + \mathbf{v}, \mathbf{u} - \mathbf{v} \rangle = \|\mathbf{u}\|^2 - \|\mathbf{v}\|^2$

(e) $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + 2 \langle \mathbf{u}, \mathbf{v} \rangle + \|\mathbf{v}\|^2$